



CHAPTER ATA-56 WINDOWS

THIS MANUAL IS INTENDED FOR TRAINING PURPOSE ONLY

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Windows, General

Introduction

The flight compartment windows give the flight crew an adequate, unobstructed view outside the aircraft under all flight and weather conditions. The windows are structural elements which are able to withstand pressurization forces and impact from a bird strike.

The passenger compartment windows give the passengers a view outside the aircraft. The windows are able to withstand pressurization forces and insulate the passengers from the extreme temperatures outside the aircraft.

General Description

[Refer Figure 56-1](#) & [Refer Figure 56-2](#)

The flight compartment has a pilot's windshield and side window, and a copilot's windshield and side window.

The passenger compartment has 29 window assemblies on the left side and 26 window assemblies on the right side.

On aircraft with ModSum 4-190561 incorporated, additional 3 window assemblies are installed on the right side in lieu of type II/III emergency exit. One window assembly is installed in the emergency exit door, aft passenger door, and aft service door.

Detailed Description

Each passenger compartment has an outer and inner acrylic panel separated around the peripheral edge by a polycarbonate spacer and a neoprene gasket, bonded to the outer panel.

The assembly is put on a silicone rubber seal, and is installed in an aluminum alloy window frame by eight retaining clips that are bolted to the frame. The space between the panels is vented to the passenger compartment ambient by two holes in the inner panel.

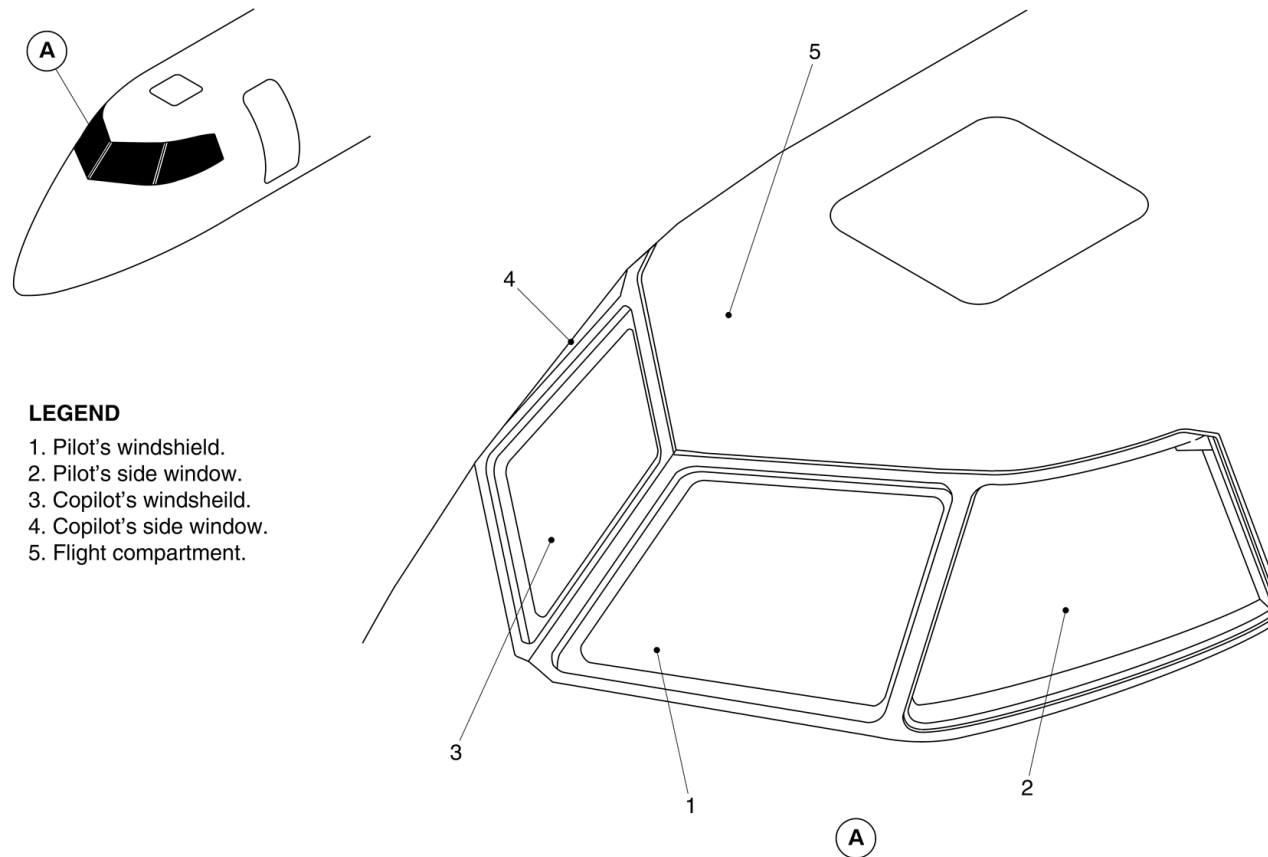


Figure 56-1 Windows – Flight Compartment

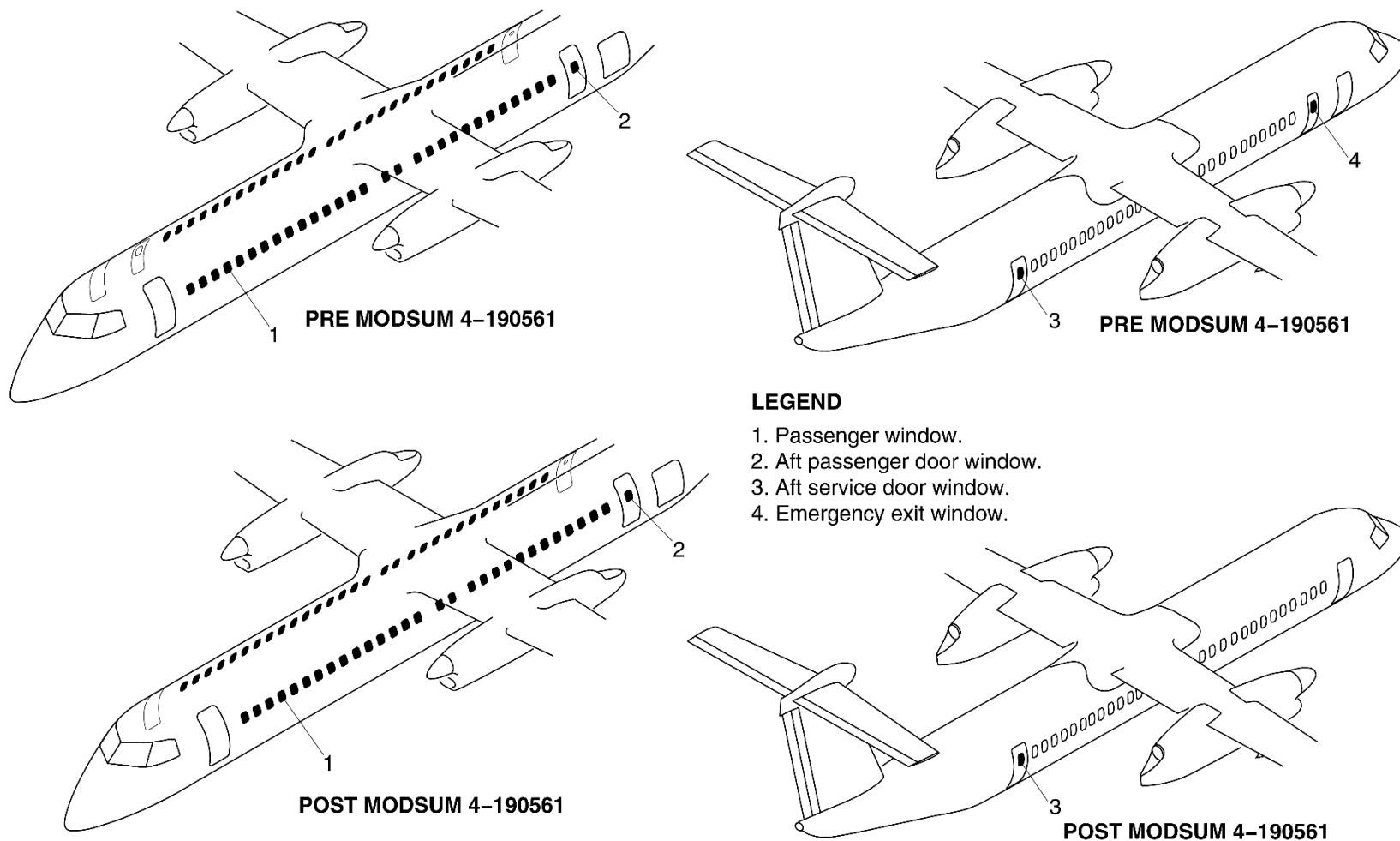


Figure 56-2 Windows – Passenger Compartment

Flight Compartment

Introduction

The flight compartment windows give the flight crew an adequate, unobstructed view outside the aircraft under all flight and weather conditions. The windows are structural elements which are able to withstand pressurization forces and impact from a bird strike.

General Description

[Refer Figure 56-3](#)

The windows in the flight compartment consist of a pilot's windshield and side window and a copilot's windshield and side window. The windshields are a fail-safe glass design with an anti-ice heating system powered by the 115 Vac electrical system. The side windows are a fail-safe acrylic design. Only the pilot's side window has 115 Vac anti-ice heating.

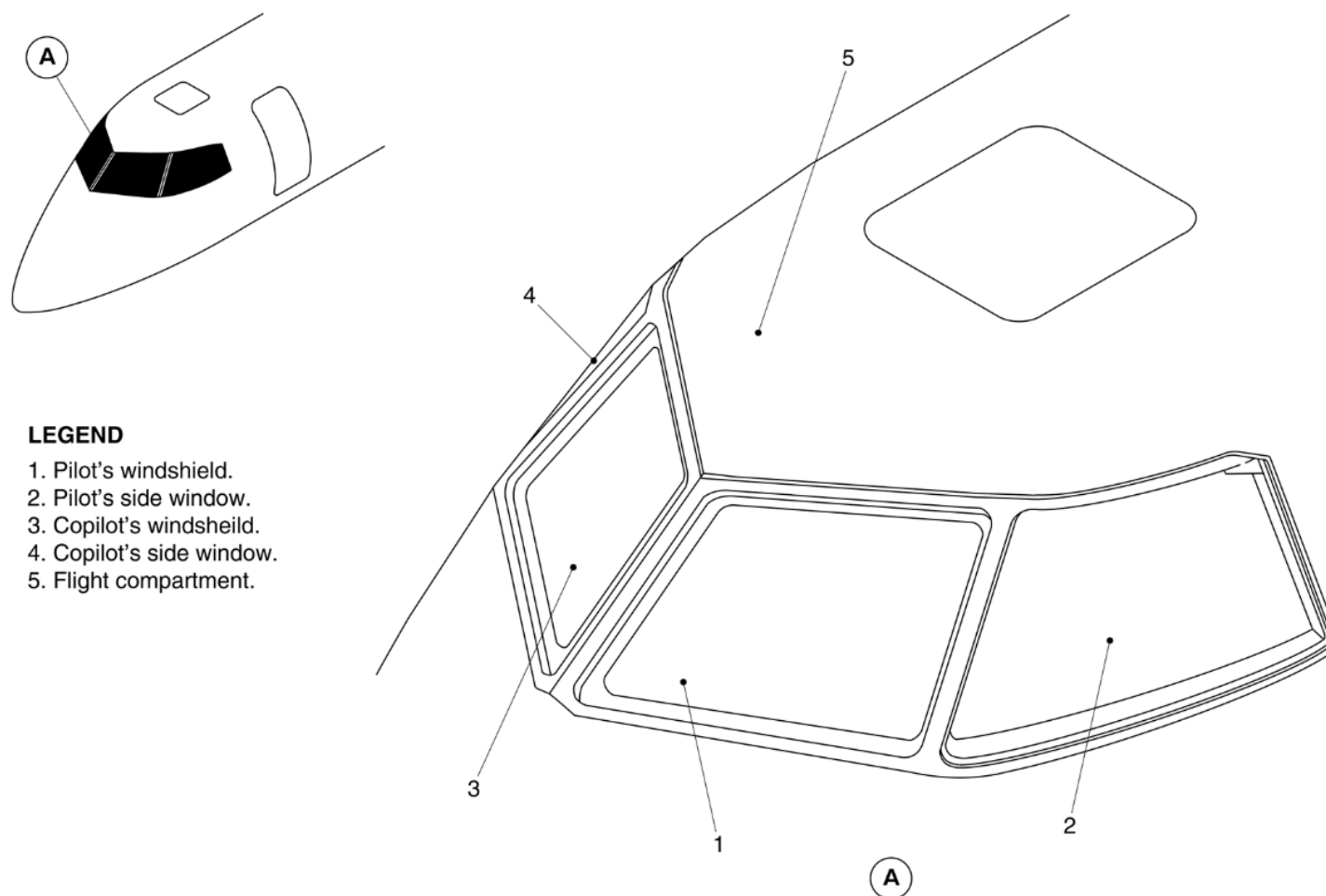


Figure 56-3 Flight Compartment Windows - Location

Detailed Description

[Refer Figure 56-4](#) & [Refer Figure 56-5](#)

The flight compartment has two main windshields, one for the pilot and one for the copilot. Each windshield has a 0.120 in. (3.05 mm) thermally tempered glass outer face ply coated on the inside with a conductive, metallic film which supplies heating for anti-ice purposes. The heating film extends up to 0.25 in. (6.35 mm) from the edge of the face ply and covers most of the windshield area except for the extreme forward section.

A heating element terminal block is located near the top of the face ply on the inboard side. Two urethane face ply interlayers are adjacent to the outer glass face ply. Three temperature sensors are imbedded in the interlayer immediately next to the glass face ply.

The main structural elements of the windshield are two chemically strengthened glass main plies. The outer mainly has a thickness of 0.230 in. (5.84 mm) while the inner main ply has a thickness of 0.180 in. (4.57 mm). The two main plies are separated by a vinyl interlayer.

A molded, silicone rubber gasket is bonded around the edge of the windshield and serves as a weather and pressure seal. The windshield is secured to the airframe by three external, aluminum alloy retainers with bolts. The rear edge of the window is secured by the side window retainer.

The copilot's side window has two structural stretched acrylic mainplies 0.38 and 0.31 in. (9.65 and 7.87 mm) thickness, laminated together with a vinyl interlayer.

There is a polysulfide weather seal and a molded rubber pressure seal around the edge of the window. The window is retained by aluminum retainers and bolts.

The pilot's side window is similar to the copilot's side window but has an outer, non-structural, acrylic faceply of 0.1 in. (2.54 mm) thickness. The faceply is coated on the inside with a conductive, metallic film which supplies

heating for anti-ice purposes.

The heating element terminal block is made of Diallyl Phthalate. It has connections for three temperature sensors and located near the top of the faceply on the inboard side. It has eight connections, in that two pairs of connections are used for normal and overheat sensors.

One pair for the power supply to the heaters. The fourth pair of connections are installed with a spare sensor. Only the forward part of the pilots side window is heated.

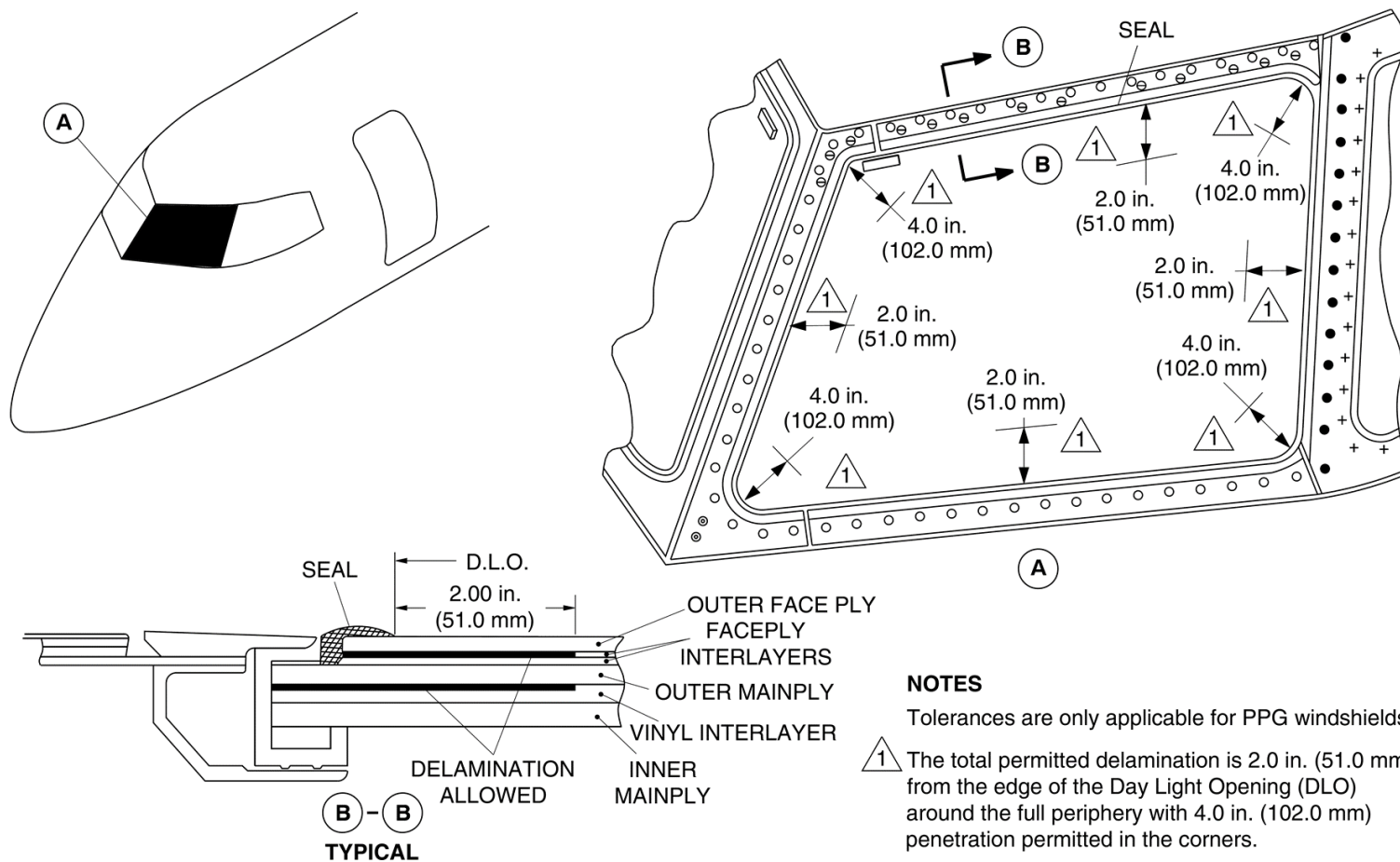


Figure 56-4 Pilot's Windshield - Cross-Section

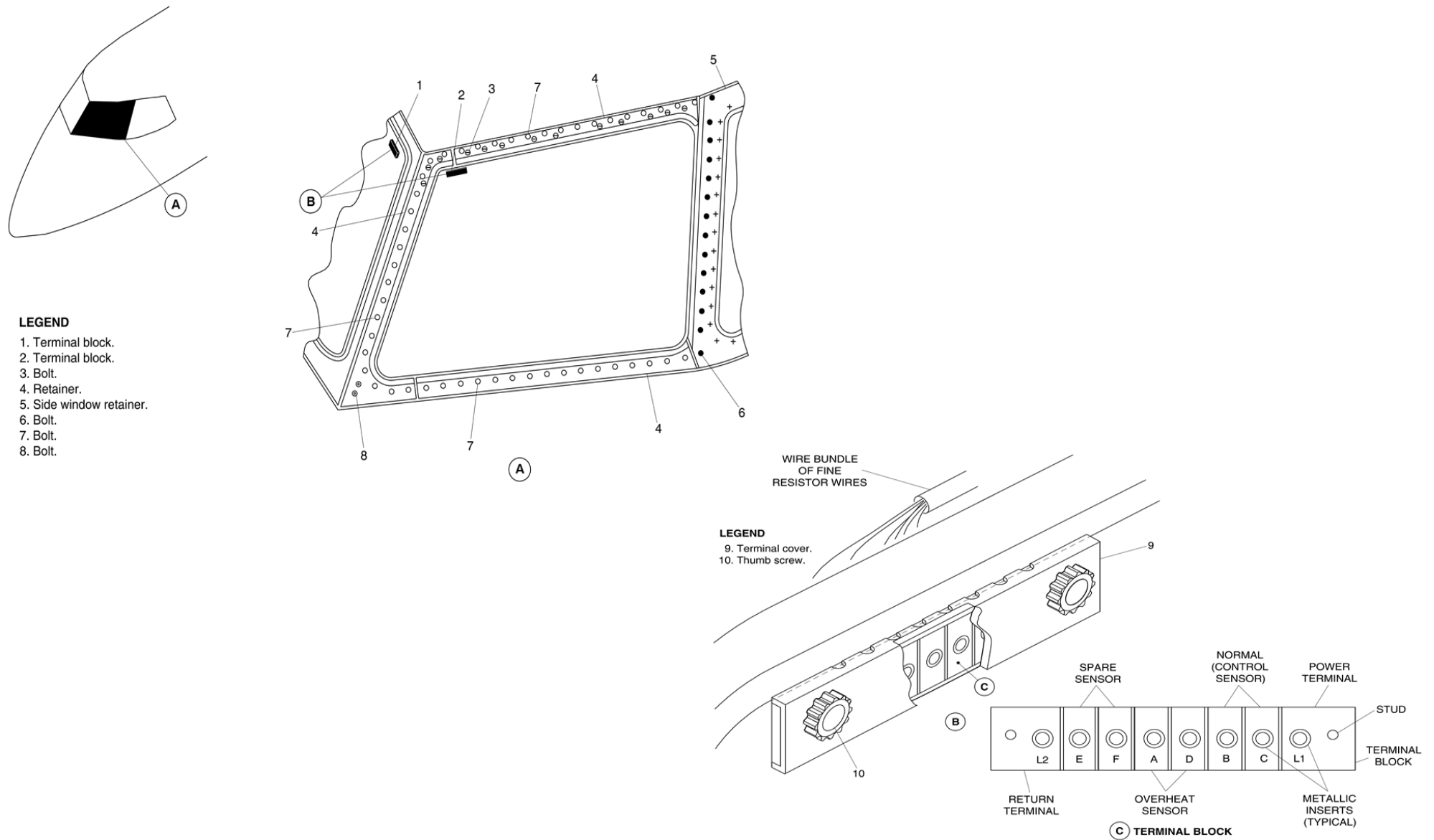


Figure 56-5 Windshield

Passenger Compartment

Introduction

The passenger compartment windows give the passengers a view outside the aircraft. The windows are able to withstand pressurization forces and insulate the passengers from the extreme temperatures outside the aircraft.

General Description

[Refer Figure 56-6](#)

There are 29 window assemblies installed in the left side of the passenger compartment and 26 window assemblies installed in the right side of the passenger compartment. On aircraft with ModSum 4-190561 incorporated, additional 3 window assemblies are installed in lieu of type II/III emergency exit door. A window assembly is installed in the emergency exit door, aft passenger door, and aft service door.

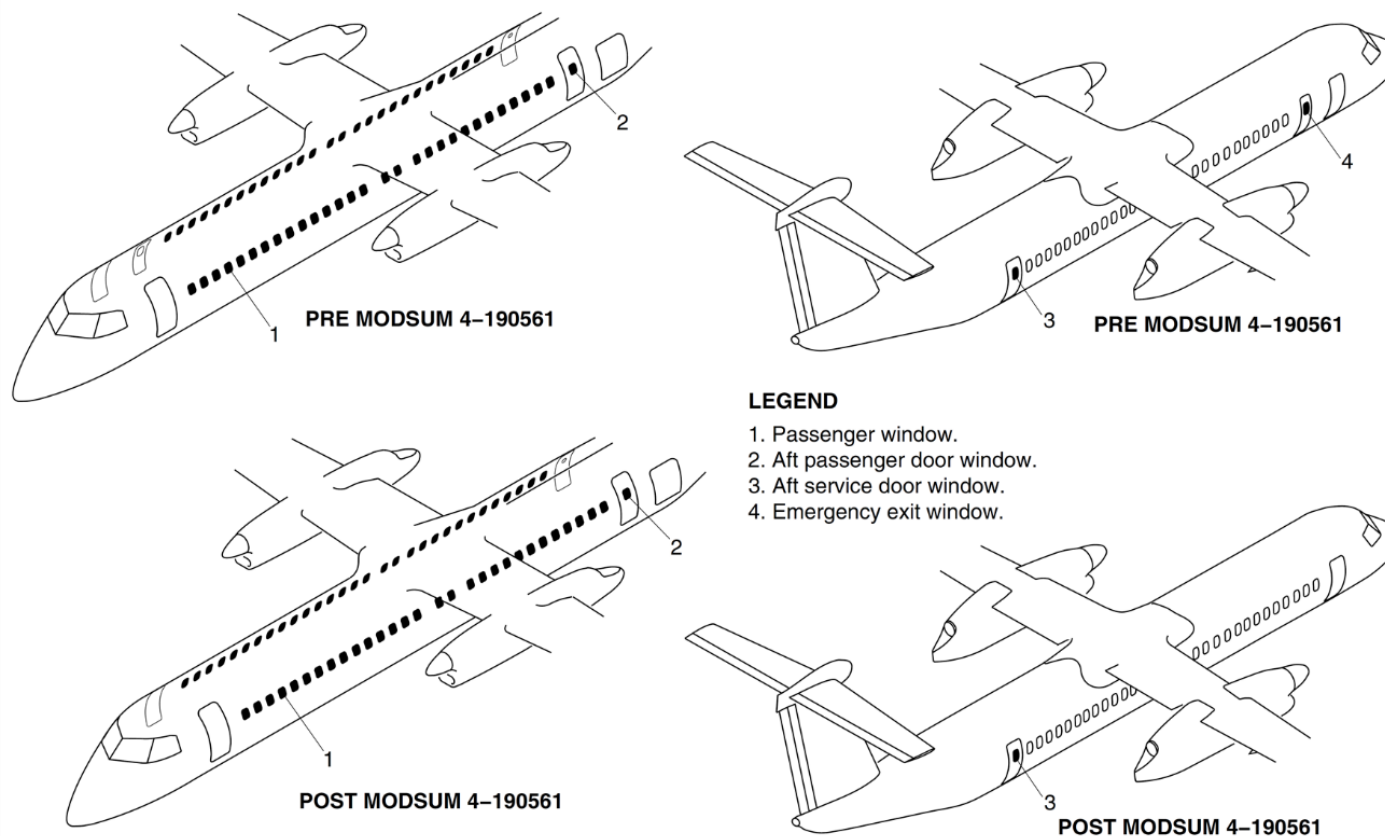


Figure 56-6 Flight Compartment Windows - Location

Detailed Description

[Refer Figure 56-7](#)

Each passenger compartment window has an outer and inner acrylic panel separated around the edge by a polycarbonate spacer and a neoprene gasket bonded to the outer panel.

The assembly is fitted with a silicone rubber seal and installed in an aluminum alloy window frame by eight retaining clips that are bolted to the frame. The space between the panels is vented to the passenger compartment ambient air by two holes in the inner panel. The window in the emergency exit door is of similar construction.

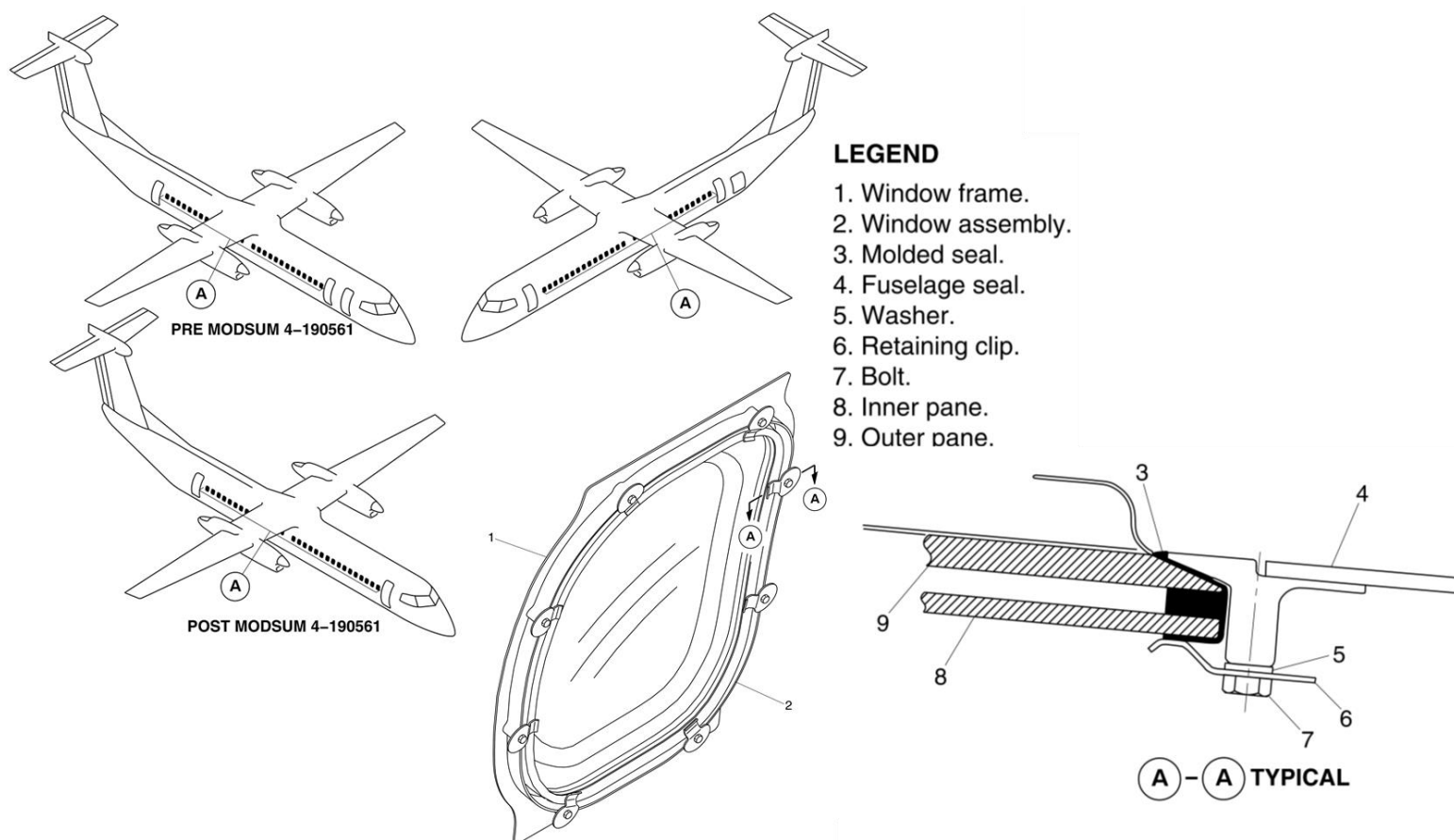


Figure 56-7 Passenger Compartment Windows

Refer Figure 56-8

The window in the aft passenger door has an outer and inner acrylic panel separated around the edge by a polycarbonate spacer and a neoprene gasket. The gasket is bonded to the outer panel. The assembly is fitted with a silicone rubber seal and installed in an aluminum alloy window frame by **four retaining**

clips that are bolted to the frame. The space between the panels is vented to the passenger compartment ambient air by two holes in the inner panel. The window in the aft service door is of similar construction.

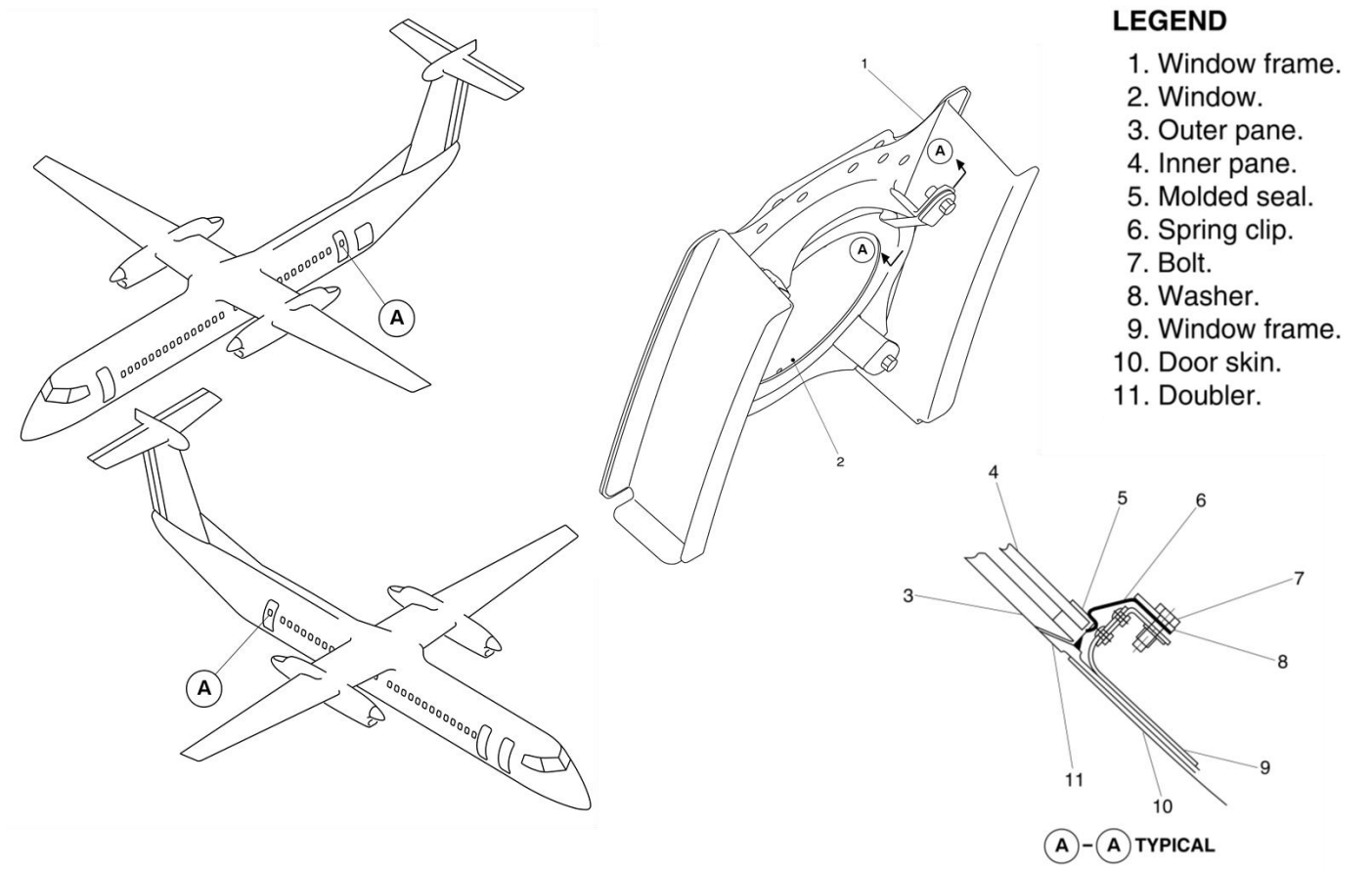


Figure 56-8 Passenger Compartment Door Windows